



Drone Legislation

If you own a drone and are worried about restrictive legislations, you're in for a treat. <http://dgit.in/DroneLegis>



Big bad telescope

Nasa is going to launch another space telescope much more powerful than the Hubble. <http://dgit.in/BigTeles>



One ARM to rule them all?

Championing the cause of RISC like no other, ARM's finding itself at the centre of almost all emerging hardware technology trends of the 21st century. We caught up with Graham Budd, Chief Operating Officer, ARM, to get a sense of what's keeping them busy.

◀ Graham Budd,
Chief Operating Officer, ARM

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Q ARM only offers processor design and not necessarily the SoC as a complete platform. Why?

Graham: Actually, ARM also provides interconnects in specific system IP components and other types of IP that we create will help SoC designers to put these chips together more quickly and efficiently in a physical IP. The cell libraries for any SoC design are also shared so that a company doesn't have to reinvent the wheel. This is one of the strengths, I think, of ARM's business model because we are enabling many different companies to make their own systems on chip using our IPs in their own different ways. So there is a lot of healthy competition in the market that supports innovation and it is definitely one of the strong factors that has helped us in contributing towards the amazing innovation that we saw over the last few years, for example in the smartphone or tablet market.

Q A lot of companies, including yours, are trying to tackle the power efficiency problem at the hardware level during the CPU design stage. Does ARM also look at the power problem from another perspective, like, thinking beyond Lithium-ion or Li-po batteries?

Graham: So we think about power from a perspective of the whole system even though we are not providing all the components of the system. For ARM, power efficiency has been in the DNA of this company since the very start when ARM was founded 25 years ago. And it has always been at it from a system context but also it's about the scalability of the architecture. This is so important because having that broad range of different power-performance processors enables us to use the same architecture. At one extreme end we have the very tiny power sipping cortex M0 designed to enable end-point IoTs that could run on tiny Lithium-ion batteries for years, and on the other end we have efficient data centre servers that are obviously consuming more power than the M0 but are still optimised for the type of workloads that are being run in that data centre. And it's that scalability across the breadth of usage scenarios that really helps as well. So power means three things for me – (1) It's the inherent power efficiency of the design which we are always concerned about, (2) the power efficiency of the system and then (3) the scalability of the architecture that enables people to build efficient systems.

Q Internet of Things, connected cars, how is ARM at the centre of all these innovative technology trends?

Graham: The connected car space is a very exciting area for ARM and we are seeing a very rapid growth in advanced technology being built into vehicles, particularly, with driver assistance. Actually this is where the car and the IoT come together because the ADAS is going to require communication between vehicle and sensors while on the road and also for inter vehicle communication. So ARM, today, is used all over the vehicle. The Cortex M class processors are embedded into many of the electronics systems that control aspects of the chassis, braking, air-bags, etc. I think what is more exciting in the car is the developments of these new ADAS systems. We have been working on building features into our processors to make it easier for silicon suppliers and car manufacturers to build robust safety features into their cars as they get more and more connected to manage all of that complexity. So this is an exciting growth area for us, but as mentioned earlier, it is a relatively slow burn. The products that we license to companies today is going to take quite a while before they are road ready. The automotive industry is being disrupted by EV Technology from companies like Tesla, for example, are bringing down the innovation time in the industry quite rapidly and is going through quite a bit of shake up and transition at the moment and there are exciting opportunities for all. **Q**